

Utkarsh Singh

Software engineer

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Professional experience

Jaguar Land Rover Software Engineer

Jul 2022 – Oct 2025, Bengaluru

Vehicle network simulator

- Developed the in-house simulator in **Python** with a 3-engineer team, replacing Intrepid, Vector & dSPACE tool licences worth a projected **£500k**.
- Developed desktop, headless & touchscreen versions using **PySide6, Flask & reTerminal** (Raspberry Pi), used on **500+ machines** per telemetry.
- Built XML parser using **ElementTree** by reverse engineering the undocumented **100k-line spec**, rarely needing changes for quarterly releases.
- Implemented a **C++** port of the parser using pugixml & std::map of typed structs for real-time margins, cutting the parse time to **0.1s from 2-3s**.
- Solely owned the signal encoding, decoding & transmission end-to-end using **scapy & threading**, shipping **20+ features** such as fault injection.
- Built dashboard builder in **PySide6** with custom **drag-and-drop** widgets & per-widget code editor, exporting dashboards as a standalone .py file.
- Developed the touchscreen app for **100+ Apple CarPlay lab rigs** using **reTerminal**, with 10 signals such as vehicle speed, gear, wakeup & sleep.

Data engineering

- Built **Jira-to-BigQuery** data pipelines for **25+ projects** using Python's jira library, Cloud Functions & Cloud Scheduler, running 2-hourly to weekly.
- Designed **40+ Tableau dashboards** reading from BigQuery for management to self-serve, ranked **#1** among all creators org-wide based on usage.
- Developed **Selenium** scraper to extract XML data from 20+ internal REST endpoints into BigQuery tables covering all **40k** JLR employee records.

Workflow automation

- Built **Excel-to-Jira** automation in PySide6 for weekly supplier sheets, creating **1000s of tickets** & colour-coding mismatches in the sheet itself.

Independent projects

gitsearch [demo](#), [code](#)

Apr – Aug 2026

- Developed semantic search over **244k GitHub repos** with 200+ stars, using Python, FastAPI & **Postgres + pgvector**, finding them by what they do.
- Utilised GitHub GraphQL & REST APIs to crawl **267k repos** & index 244k with **bge-small** embeddings, with weekly refresh jobs on GitHub Actions.
- Implemented **hybrid retrieval** using Postgres full text, pgvector & pg_trgm in a **single SQL query**, with a breakdown of why each result ranked.
- Built an AI-powered guide agent using **Claude tool calls** over the repo files, allowing users to get a step-by-step usage guide for any search result.
- Utilised Supabase, GCP Cloud Run & GitHub Pages to host the full stack under **\$30/month** with scale-to-zero, serving warm searches in **120ms**.

The Good Guest [site](#), [code](#)

Jan – Aug 2026

- Designed & developed a product spanning an **iOS capture app, GCP backend & 3D web viewer**, letting users scan a room & redesign it over chat.
- Built capture flow in **SwiftUI & ARKit** with **RoomPlan** for room geometry, uploading the LiDAR scans to Google Cloud Storage in the background.
- Utilised Meta's **SAM 3 & SAM 3D Objects** for detecting & rebuilding furniture in 3D such as sofas, beds & tables in FastAPI on Cloud Run L4 GPUs.
- Developed a 3D room viewer in **Next.js, three.js & WebGL2** for exploring the scanned rooms in any browser, with **5.8x smaller** splat files via SPZ.
- Built an AI redesign agent using **Claude tool calls** with a backend solver checking each move, allowing users to rearrange furniture by chatting.
- Utilised **Firebase Auth SDK** for Apple & Google sign-in, creating iOS flows such as Live Activity progress & account deletion, as per App Store rules.

Smaller builds

Jul – Aug 2026

- Built **Shrink to Fit** for compressing images & PDFs to exact sizes like 500KB or 2MB in the browser using canvas, pdf.js & **Ghostscript WASM**. [code](#)
- Built read-only chat viewers for **WhatsApp & Instagram** exports in **vanilla JS** with voice note waveforms, search, zip repair & **zero uploads**. [code](#)
- Built 20 generative art pieces such as harmonographs, Lissajous tables & Truchet tiles in **HTML Canvas & GLSL** for **linework**, exporting plotter SVG.

Education

Indian Institute of Technology Delhi B.Tech + M.Tech Dual Degree, Chemical Engineering, CGPA 8.6/10

Jul 2017 – May 2022

- Built **packed-bed reactor geometries** in **Blender's** rigid body physics for the thesis, with porosity analysis in **MATLAB** & CFD runs in **Ansys Fluent**.

Skills

AI engineering

AI Agents, Tool Use, Anthropic API, Gemini API, Embeddings, pgvector, Hybrid Search, LLM Evals, Gaussian Splatting

Backend

Python, C++, FastAPI, Flask, PostgreSQL, SQL, Protobuf, scapy, Multithreading, Docker, GitHub Actions

Frontend & mobile

Swift, SwiftUI, ARKit, RoomPlan, Next.js, React, TypeScript, JavaScript, three.js, WebGL, GLSL, PySide6, Qt

Cloud

GCP, Cloud Run, Cloud Functions, Cloud Scheduler, BigQuery, Firestore, Firebase, Supabase, Embedded Linux

Data

BigQuery, Tableau, pandas, Selenium, Jira

Honors & awards

- Exceptional Creator Award, Jaguar Land Rover, 2025
- Department Rank 1, Chemical Engineering Dual Degree batch, IIT Delhi, 2022
- Semester Merit Award ×2, IIT Delhi
- JEE Advanced AIR 2525 (top 1.14%), JEE Main AIR 742 (top 0.06%), 2017
- SOF Mathematics Olympiad, Zonal Rank 3 (finals); Rank 3, inter-school mathematics exam, awarded by the Governor of Uttar Pradesh